

A/B Testing for Oil Well Production

Example comparison of two IP30 prediction models

Based on the uploaded notebook `main.ipynb`

Business question

Does the new production model predict significantly better oil well IP30 performance than the old standard model for the same wells?

Test design from the notebook

- **A group:** Model A, the old standard model
- **B group:** Model B, the new proposed model
- **Sample:** 50 fictitious wells in one region
- **Input feature:** permeability sampled from 10 to 200 mD
- **Reproducibility:** random seed fixed at **42**
- **Statistical test:** paired t-test at significance level $\alpha = 0.05$

Model formulas used

Model A: $IP30 = 0.5 \times \text{permeability} + \text{Normal}(0, 5)$

Model B: $IP30 = 0.7 \times \text{permeability} + \text{Normal}(0, 3)$

Interpretation: Model B is designed to produce higher expected IP30 and lower noise for the same permeability values.

50

WELLS ANALYZED

10–200 mD

PERMEABILITY RANGE

Paired t-test

PRIMARY HYPOTHESIS TEST

Seed = 42

REPRODUCIBLE OUTPUT

Notebook also generated histograms, density curves, scatter plots, box plots, and the distribution of paired differences to visually compare the two models.

Results and Decision

Model B outperformed Model A in this oil production example

Metric	Model A (Old)	Model B (New)
Mean IP30 (bbl/day)	47.35	66.14
Variance	756.55	1532.95
Standard deviation	27.51	39.15
Sample size	50	50
Mean improvement (B – A)	18.79 bbl/day	
Paired t-statistic	10.0223	
Paired p-value	0.0000 (< 0.05)	

Decision: Reject H0. The notebook concludes that Model B performs significantly better than Model A, and the improvement is unlikely to be due to chance.

What the plots show

- Histogram and PDF curves shift to the right for Model B
- Scatter plot shows higher predicted IP30 for the same permeability values
- Box plot confirms distribution differences between models
- Paired-difference distribution centers above zero

Repository-ready takeaway

This A/B testing example demonstrates a clean, reproducible workflow for comparing two oil production models. Using the same wells and a paired t-test, the new model delivers a materially higher average IP30 and passes the significance threshold.

Recommended caption: “A/B test of oil well IP30 models shows the proposed Model B significantly outperforms the legacy Model A under a reproducible simulated dataset.”

Source values taken directly from **main.ipynb**: mean A = 47.35, mean B = 66.14, paired t = 10.0223, p-value = 0.0000, random seed = 42.